

1. (5 pontos) Based on Kurlat (2020, Cap. 6)

Suppose a household solves the following two-period consumption-savings problem with taxes:

$$\begin{aligned} \max_{\{c_1, a, c_2\}} \quad & u(c_1) + \beta u(c_2) \\ \text{s.t.} \quad & \\ & c_1 + a = a_0 + y_1 - \tau_1 \\ & c_2 = y_2 - \tau_2 + (1+r)a \end{aligned}$$

with $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$ for $\sigma \neq 1$ and $u(c) = \ln c$ for $\sigma = 1$, where: c_1, y_1 and τ_1 are consumption, income and taxes at period 1, c_2, y_2 and τ_2 are consumption, income and taxes at period 2, a_0 is an exogenous initial level of wealth, and r is the real interest rate at which the household can freely borrow and lend. Assume $\beta \in (0, 1)$, $\sigma > 0$ and $r > -1$.

- Solve for the household's choice of c_1, c_2 and a in closed form.
- How does $\frac{c_1}{y_1}$ depend on y_2 ? How can we relate this to households suddenly becoming more optimistic about the future?
- Suppose $y_2 = \tau_1 = \tau_2 = 0$ and compute $\frac{\partial c_1}{\partial r}$. How does the answer depend on σ ? Interpret the answer.
Hint: Think about what σ means for the relative importance of income and substitution effects
- Return to the general case and consider the effects of taxation. Show how consumption in both periods and savings between period one and two are affected by government taxation.
- Show that τ_1 and τ_2 affect consumption only through the present value of $\tau_1 + \frac{\tau_2}{1+r}$. Using this, describe what happens to c_1, c_2 and a if the government cuts τ_1 by one unit and raises τ_2 by $1+r$. Does the timing of the taxation matter for households' consumption?

2. Based on Kurlat (2020, Cap. 6)

Suppose a household solves a variant of the previous problem

$$\begin{aligned} \max_{\{c_1, a, c_2\}} \quad & u(c_1) + \beta u(c_2) \\ \text{s.t.} \quad & \\ & c_1 + a = y_1 - \tau_1 \\ & c_2 = y_2 - \tau_2 + (1+r)a \\ & a \geq -b \end{aligned}$$

- What does the new constraint mean? What does b represent?
- Give two examples of when this constraint is more likely to be binding
- Solve for c_1, c_2 and a .

Hint: Notice the new constraint may or may not be binding. Consider both cases

- d) Suppose now that, rather than levying a lump-sum tax in period 2, the government taxes the return to savings. The second-period budget constraint becomes

$$c_2 = y_2 + (1 + r - \tau_2)a$$

Compare this tax with a lump-sum tax that raises the same revenue. Show that taxing savings will change the ratio $\frac{c_2}{c_1}$ while a lump-sum tax does not, and explain which margin each type of taxation distorts.